RAI Risk Taxonomy 2.0

Current hierarchy, reproducible mapping, and reliability validation

RAI Risk Taxonomy Project

25 July 2026 · release candidate v2.18.0-rc

Abstract

RAI Risk Taxonomy 2.0 organizes responsible AI risks across General-purpose AI, Agentic AI, and Physical AI. The current release preserves 1,711 registered identifiers and distinguishes 1,660 independent active L4 cards from 51 merged provenance records. A hybrid design combines standards-led hierarchy construction with literature-derived L4 evidence. The current hierarchy contains three semantic L2 categories and 54 semantic L3 families. Four General-purpose AI families were added for societal impact, and nine former Physical AI societal families were migrated into the shared General-purpose AI branch. L4 placement uses BGE-M3 embeddings, bilingual L3 definitions, centroid similarity, direct definition similarity, and keyword similarity under domain constraints. Reproducibility is supported by fixed inputs, model snapshot, deterministic seeds, content hashes, iteration traces, and invariant tests. On 1,660 active cards, seeded spherical EM converges in 20 iterations at a mean cosine objective of 0.8005. The released L3 appears within the nearest centroid at Top-1 for 71.9% of cards and within Top-5 for 95.2%. These results establish a reproducible current baseline while preserving human review for semantically ambiguous cases.

Keywords: AI risk taxonomy; semantic mapping; BGE-M3; reproducibility; Agentic AI; Physical AI.

1 Scope and design objective

AI risk has expanded from text-content harms to autonomous action, tool use, multi-agent interaction, and direct physical-world effects. RAI Risk Taxonomy 2.0 addresses this broader scope through a five-level structure from L0 to L4. L1 separates three technological domains. L2 captures the common safety dimension. L3 defines risk families. L4 records a concrete risk mechanism with a stable identifier, bilingual label and definition, metrics, and evidence.

The report distinguishes identifiers from active analytical units. The registry contains 1,711 stable IDs. Of these, 1,660 are independent active cards and 51 are merged records retained for citation and provenance. All hierarchy statistics and reliability tests use the 1,660 active cards unless a table explicitly states otherwise.

2 Evidence and registry construction

2.1 Top-down evidence filter

The integrated evidence space contains papers, policy reports, and patents. AI-risk evidence is selected through reviewed risk-source collections, record-type inclusion rules, valid-title checks, schema standardization, and within-type deduplication. Evidence records support risk interpretation but do not map one-to-one onto L4 cards.

Table 1: Evidence-to-taxonomy stages in the current release.

Stage	Count	Unit	Inclusion rule
AI evidence	3,906,767	Record	AI source collection, valid title, standardized schema, within-type deduplication
AI-risk evidence	82,971	Record	Reviewed AI-risk sources and record-type filters
Exact-deduplicated candidates	1,725	Card candidate	Exact normalized label and core-definition match
Registered L4 IDs	1,711	Identifier	L3-label leakage removed while identifiers remain auditable
Active L4 cards	1,660	Independent card	Registry-preserving consolidation excludes 51 merged records from active analysis

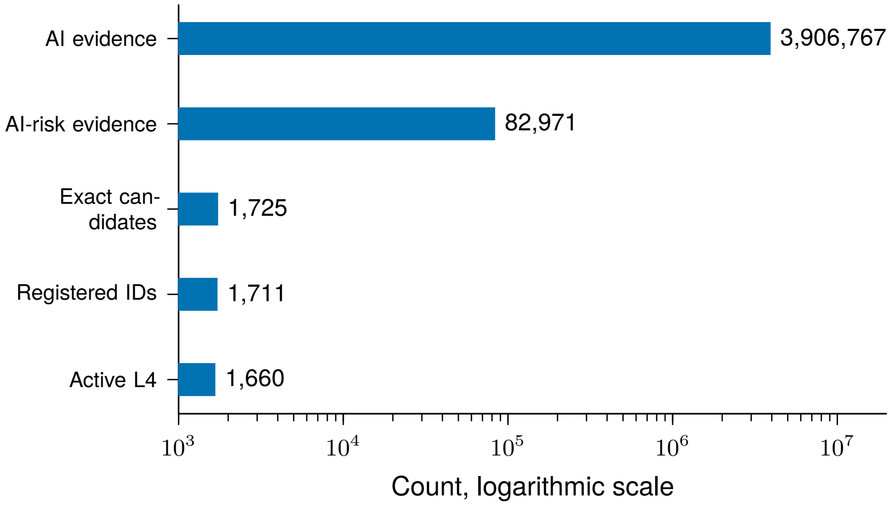

Figure 1: **Evidence and registry funnel.** The statistical unit changes from evidence records to risk-card identifiers. Active analysis excludes merged records without deleting their IDs.

2.2 Registry invariants

The current build enforces four invariants:

1. every registered identifier occurs exactly once;
2. every merged record points to an active canonical card;
3. every active card has one operational primary path and one semantic L3 path;
4. retired or merged records are excluded from active-card metrics.

3 Current hierarchy

3.1 L1 to L3 structure

The semantic L2 vocabulary is Interaction Safety, System Safety, and Societal Impact. These categories are instantiated only where a domain has relevant families. The current hierarchy contains 54 semantic L3 families: 33 General, 6 Agentic, and 15 Physical. Four General societal families were added: Energy Consumption and Environmental Pollution, Lack of Accountability and Governance Framework, Lack of Transparency, and Fairness. Nine former Physical societal families were migrated into General Societal Impact. Existing similarly named families were retained so that L4 cards could be assigned by relative semantic fit rather than by label deletion.

Table 2: Current hierarchy coverage by L1 domain.

L1 domain	Semantic L3	Active primary-path L4	Share of active L4
General-purpose AI	33	1,221	73.6%
Agentic AI	6	281	16.9%
Physical AI	15	158	9.5%
Total	54	1,660	100.0%

The primary-path counts in Table 2 describe the deployed navigation tree. For semantic analysis, cards under review are attributed to their retained semantic review path. This produces the L1 by L2 distribution in Table 3. The five-card difference between primary-path and semantic domain counts arises from cross-domain review paths and does not change the total.

Table 3: Active L4 cards by semantic L1 and L2 path.

L1 domain	Interaction Safety	System Safety	Societal Impact	Total
General-purpose AI	641	467	108	1,216
Agentic AI	0	286	0	286
Physical AI	64	94	0	158
Total	705	847	108	1,660

3.2 Hierarchy coverage

All 1,660 active cards have a semantic L3 path. All 54 semantic L3 families are occupied and each contains at least one released card outside the review subset. Anthropomorphism is the largest family with 160 cards, equal to 9.6% of active L4 cards. The five largest families contain 553 cards, or 33.3%.

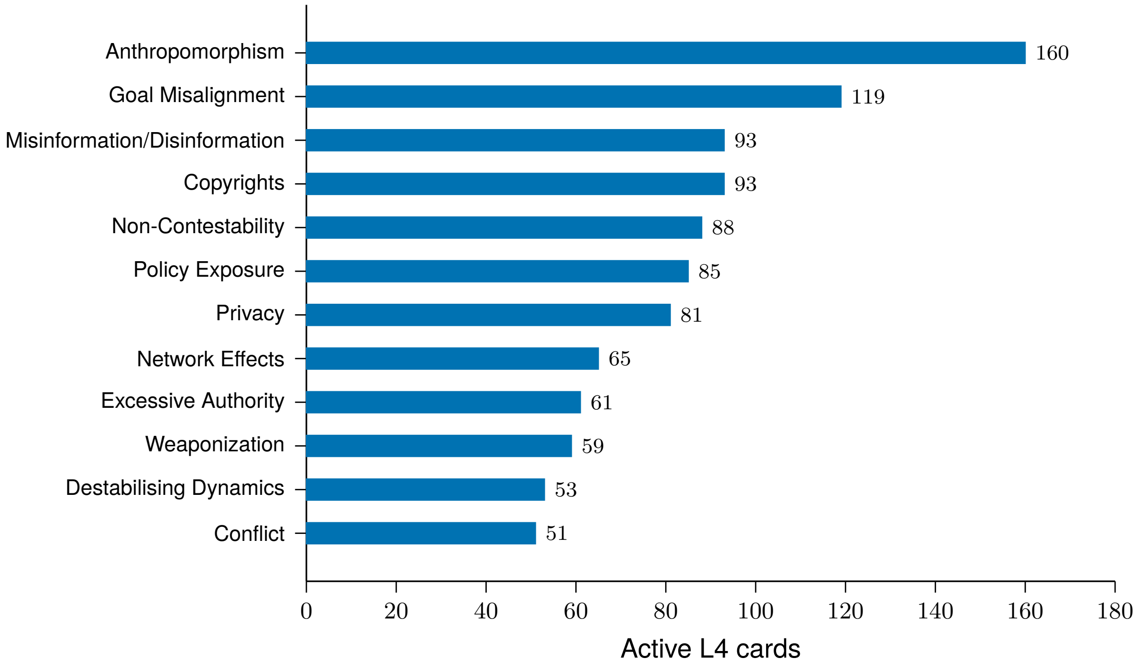

Figure 2: **Twelve largest semantic L3 families.** Counts use the retained semantic path for boundary-review cards and exclude 51 merged records.

4 Reproducible L4 to L3 mapping

4.1 Representations and constraints

Each active card is represented by the concatenation of its English and Korean label and definition. Each L3 seed uses the same four fields. BGE-M3 encodes both card and seed text into normalized 1,024-dimensional vectors. English TF-IDF vectors provide a complementary keyword signal.

Physical source-card identities remain locked, and their paths are changed only when an approved revision is imported from the authoritative Physical AI master. The 31 July 2026 synchronization imported two such revisions after 24-condition EM sensitivity testing and 1,000 bootstrap repeats. General cards can compete across General families. Agentic candidates require an explicit autonomous execution mechanism unless the source card is already Agentic. The migrated societal families compete inside the General branch. These constraints prevent generic vocabulary from moving cards into a technology-specific domain.

4.2 Iterative winner assignment

For card i and candidate family k, the mapping score is

$$S_{ik} = 0.60 \cos(x_i, \mu_k) + 0.30 \cos(x_i, s_k) + 0.10 \cos(t_i, q_k), \tag{1}$$

where x_i is the normalized BGE-M3 card embedding, μ_k is the current family centroid, s_k is the bilingual L3 seed embedding, and t_i and q_k are TF-IDF representations. Centroids are updated from assigned members, definition seeds, and approved prior members:

$$\mu_k \leftarrow \text{norm}\left(0.75\bar{x}_k + 0.15 s_k + 0.10\bar{x}_k^{prior}\right). \tag{2}$$

When no prior member set exists, the member and seed weights are 0.85 and 0.15. The constrained winner is accepted only when it improves the current score by the configured margin. The release build reached a fixed point after four mapping iterations, with 144, 13, 5, and 0 changed cards. In total, 182 active cards received a structural or competitive path update, including 30 societal structure migrations.

4.3 Boundary-review status

The review marker is metadata on a semantic assignment, not a separate risk concept. It is retained when a proposed placement does not jointly satisfy a direct seed cosine of at least 0.45 and a composite-score margin of at least 0.015, or when the card lies outside the approved extension scope. The current release resolves 151 prior review markers and retains 614 cards for subsequent human adjudication. This status is reported only as a quality-control boundary and is not counted as an L2 or L3 family.

5 Reliability validation

5.1 Validation design

The released semantic path is evaluated against centroids formed from the released assignments. Three diagnostics are used:

1. **EM convergence**: seeded spherical EM repeatedly updates centroids and assignments until no assignment changes.
2. **Top-k containment**: the share of cards whose released L3 appears among the k nearest released-family centroids.
3. **Perturbation stability**: agreement between unperturbed and perturbed nearest-centroid assignments after Gaussian noise is added to normalized embeddings.

The validation uses seed 20260723, 5,000 label permutations, and 200 perturbation repeats at each noise level. All 54 L3 seed rows are retained in both evaluated populations. The released subset excludes 614 boundary-review cards but does not refit or reduce the L3 vocabulary.

Table 4: Current reliability results at v2.18.0-rc.

Diagnostic	All active cards	Released subset
Cards assessed	1,660	1,046
Semantic L3 families	54	54
EM iterations to fixed point	20	12
Final mean cosine objective	0.8005	0.8026
Top-1 containment	71.9%	81.6%
Top-2 containment	85.3%	91.2%
Top-3 containment	91.3%	94.6%
Top-5 containment	95.2%	97.2%
Median assignment margin	0.0168	0.0310
Negative-margin share	28.1%	18.4%
Permutation p	0.0002	0.0002
Stability at $\sigma = 0.01$	95.0%	96.0%
Stability at $\sigma = 0.05$	75.0%	80.7%

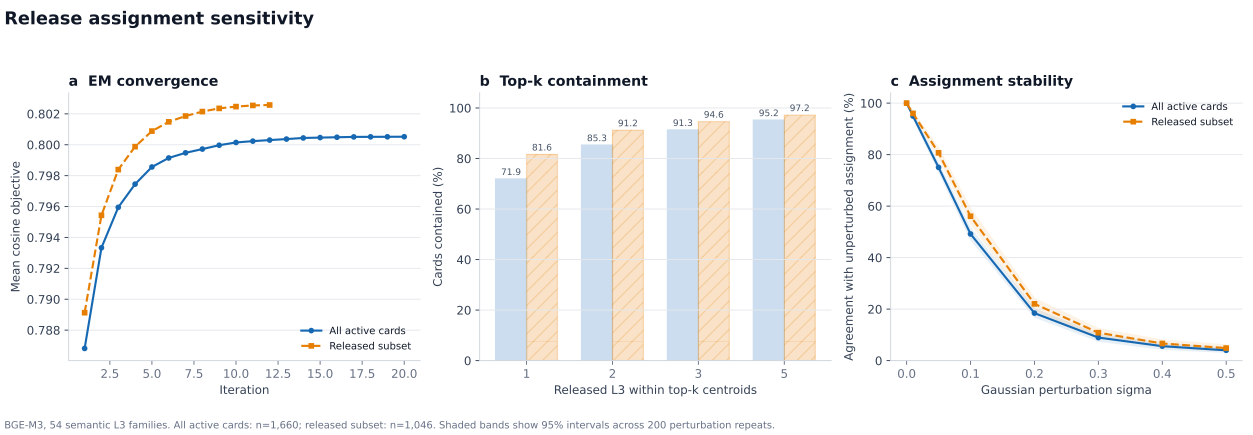

Figure 3: **Sensitivity diagnostics for all active cards and the released subset.** Panels show seeded spherical-EM convergence, Top-k containment, and assignment stability under Gaussian perturbation. Shaded bands are 95% intervals across 200 repeats.

The higher containment and perturbation stability of the released subset show that the quality-control boundary separates a less stable region of the semantic space. This comparison does not prove that every released assignment is correct. It provides a reproducible prioritization signal for human review.

6 Granularity-flow tiers

6.1 Measured merge boundary

Leaf-level granularity is fixed by a merge threshold on the cosine-similarity graph of card embeddings, where connected components are merge clusters. Rather than fixing that threshold by analyst judgment, the release derives it from the inventory. For a candidate threshold τ, let $\Phi_{\text{coh}}(\tau)$ be the pooled within-cluster mean similarity and $\Phi_{\text{att}}(\tau)$ the mean over clusters of the maximum similarity to any card outside the cluster. The operating boundary is the smallest τ at which cohesion still dominates attraction,

$$\tau^* = \min\{\tau : \Phi_{\text{coh}}(\tau) \geq \Phi_{\text{att}}(\tau)\},$$

evaluated on a 10^{-4} grid. On the normalized master inventory of 1,612 active cards, $\tau^* = 0.8329$.

6.2 Iterated consolidation

Merging at τ^* changes the density of the space, so the boundary moves with it. Repeating the cycle of merge, re-derive, and merge again generates a sequence of boundaries and inventories, the granularity flow. Each step is scored against a size-matched random-group null drawn on its own inventory; a step is admissible when at least 90% of its merge groups exceed the null by two standard deviations. Steps 1 to 5, the range released for review, satisfy this rule with a fraction of 1.00 throughout and a median z between 3.27 and 4.30. Representatives are medoids of their merge group, so every tier card is an original card rather than a synthesized summary.

Table 5: Granularity-flow tiers. Merge groups are counted per consolidation step and absorbed cards cumulatively from the Master inventory. Domain counts follow the L3 re-assignment, with the Societal Safety axis routed by scope.

Tier	τ^*	Cards	Merge groups	Absorbed	General / Agentic / Physical
Master	–	1,612	–	–	1,154 / 155 / 303
F1	0.8329	1,383	109	229	906 / 140 / 337
F2	0.8033	1,154	121	458	734 / 128 / 292
F3	0.7903	1,038	81	574	653 / 112 / 273
F4	0.7753	901	82	711	568 / 92 / 241
F5	0.7634	792	70	820	491 / 83 / 218

Societal Safety is a single concept family applied at three scopes rather than a General-only axis. Its L3 nodes therefore branch as `RAI3-{G|A|P}-SOC-nn`, where a given `nn` denotes the same risk concept at each scope; L4 identifiers are unchanged and only the owning L3 branches. A card is Physical when its label or definition carries a physical referent such as a robot, embodiment, hardware, sensor or product-safety grounding; Agentic when it is not Physical but binds to agent autonomy, long horizon goals, tool use or multi-agent interaction; and General otherwise. On the Master inventory the 430 Societal Safety cards split 351 General, 37 Agentic and 42 Physical, and the tier counts above follow that routing.

6.3 Cross-corpus replication

To test whether the procedure depends on this particular inventory, the same pipeline was applied without retuning to the MIT AI Risk Repository, whose risk categories and subcategories were embedded with the same encoder. The replication reproduces both the two transitions and the crossing, and the flow behaves as it does here.

Table 6: Granularity flow on the MIT AI Risk Repository. Merge groups and median z are per consolidation step; absorbed entries are cumulative. Domain columns are omitted because that repository uses its own top-level scheme.

Tier	τ^*	Entries	Merge groups	Absorbed	Median z	Role
Source	–	1,810	–	–	–	repository inventory
M1	0.8190	1,423	189	387	4.79	fidelity tier (crossing)
M2	0.7908	1,230	104	580	3.68	second consolidation
M3	0.7696	1,022	115	788	3.53	third consolidation
M4	0.7545	879	87	931	3.17	compression tier
M5	0.7374	727	99	1,083	3.00	extended compression tier

Every step clears the validity rule, with at least 98.9% of merge groups exceeding the size-matched random-group null by two standard deviations. The replication is reported for reference and is not released for human audit.

6.4 Released tiers

F1, F4, and F5 are released for human audit. F1 preserves the fidelity of the original inventory at the measured boundary, F4 is the compression tier reached after four consolidations, and F5 extends compression by one further step. Each tier is published as a browsable audit page in which every card is judged on description adequacy, L3 mapping, and redundancy against other cards.

7 Reproducibility controls

7.1 Fixed computational specification

Table 7: Reproducibility specification.

Component	Fixed specification
Release	v2.18.0-rc, derived from sealed v2.17.2 inputs
Encoder	BAAI/bge-m3, normalized 1,024-dimensional embeddings
Mapping seed	20260725
Validation seed	20260723
Maximum mapping iterations	30, early stop at zero changed assignments
Permutation test	5,000 matched label permutations
Perturbation test	200 repeats for $\sigma \in \{0, 0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$
Card input hash	`dff3f3269a1ed8575f409923ca0c3db1aca14931259b567abeefa12da2229acf`
Hierarchy input hash	`76156ee383cbbf84870989f2036f0a1de6f031308fd0b9f9d5f4b821f0450050`
Card-embedding hash	`8f99380ef026730105624d5103a37fea20ceb4816d2c5cace33b2e8c764c19d8`
L3-seed hash	`5ead12763d624a8e6d92dbb82536e18710d66c1670107c1d7d3c6e2ff18f6615`

7.2 Executable checks

The release build records the complete hierarchy migration, changed-card list, iteration trace, threshold decisions, and content hashes. Automated checks verify the 1,711-ID registry, 1,660 active cards, 51 merged records, 54 semantic L3 families, domain counts, review-marker count, semantic-space node count, and absence of dangling paths. Browser smoke tests cover the taxonomy explorer and semantic-space views. PDF compilation is followed by page rendering and visual inspection.

A reproduction is accepted only when the following outputs agree:

1. registry and active-card counts;
2. L1, L2, and L3 totals;
3. card and hierarchy hashes;
4. EM fixed-point iterations and final objective;
5. Top-k and perturbation metrics within the stated rounding rule.

8 Interpretation and limitations

The taxonomy is a policy-guided semantic partition, not an unsupervised claim that each risk has only one natural parent. Some L4 cards contain multiple mechanisms and can be close to several L3 families. Top-k containment is therefore reported alongside Top-1. The BGE-M3 geometry measures semantic proximity, not causal propagation, likelihood, or severity. Locked Physical paths provide a stable reference but are not an unbiased prediction set. Human evaluation remains necessary for boundary cases, newly introduced societal families, and cards affected by structural migration.

9 Conclusion

The corrected current hierarchy consists of 1,711 registered IDs, 1,660 active L4 cards, three semantic L2 categories, and 54 semantic L3 families. Its L3 composition is 33 General, 6 Agentic, and 15 Physical. Section-level tables and figures use the same active-card snapshot and explicitly distinguish primary navigation paths from retained semantic review paths. The mapping and validation workflow is reproducible from fixed model inputs, deterministic seeds, content hashes, iteration traces, and invariant tests. The current results provide a defensible baseline for human evaluation and subsequent release decisions without treating semantic similarity as ground truth.

References

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